

Grassmannian learning mutual subspace method for image set recognition

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ARTICLE INFO

Article history:

Received 21 April 2022

Revised 5 August 2022

Accepted 15 October 2022

Available online 21 October 2022

Communicated by Zidong Wang

Keywords:

Grassmannian learning mutual subspace method

Learning subspace methods

Subspace learning

Image recognition

Deep neural networks

Manifold optimization

ABSTRACT

This paper addresses the problem of object recognition given a set of images as input (e.g., multiple camera sources and video frames). Convolutional neural network (CNN)-based frameworks do not exploit these sets effectively, processing a pattern as observed, not capturing the underlying feature distribution as it does not consider the variance of images in the set. To address this issue, we propose the Grassmannian learning mutual subspace method (G-LMSM), a NN layer embedded on top of CNNs that can process image sets more effectively and can be trained in an end-to-end manner. The image set is first represented by a low-dimensional input subspace and then this input subspace is matched with dictionary subspaces by a similarity of their canonical angles, an interpretable and easy to compute metric. The key idea of G-LMSM is that the dictionary subspaces are learned as points on the Grassmann manifold, optimized with Riemannian stochastic gradient descent. This learning is stable, efficient and theoretically well-grounded. We demonstrate the effectiveness of our proposed method on hand shape recognition, face identification, and facial emotion recognition.

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1. Introduction

Multiple images of an object are useful for boosting performance of object classification [1,2]. In surveillance and industrial systems, distributed cameras and sensors capture data in a continuous stream to provide multiple images pointing to a target object. The task of object recognition based on a set of images, called *image set recognition*, is formulated to classify an image set through comparing the input set to reference image sets (dictionary) of class objects.

The image set we consider in this paper refers to a collection of image samples captured from the same object or event, thereby freeing images in a set from any order, such as temporal ordering. For example, image sets of our interest include multiview images from multiple camera sources and video frames. As in the other image classification tasks, each image in the set could be represented by a naive image vector [3–5] or by the more sophisticated feature vector via CNNs [6,7], to provide us with a set of feature vectors for set classification. Therefore, the primal issue to be addressed in the set recognition is how to represent the set which is composed of variable number of feature vectors.

Subspace-based methods [8–11] have been one of the leading solutions for image set recognition, and a central research topic for object recognition in computer vision. In the subspace methods, a set of images is effectively modeled by a low-dimensional subspace in a high-dimensional vector space [1,2,4,12]. Such subspaces are typically generated by applying either singular value decomposition (SVD) or Gram–Schmidt orthogonalization to feature vectors in a set. The subspace-based set representation is effective in the following points. (1) A subspace is invariant to the choice of basis; although many bases span the same subspace, it is unique and does not depend on the basis. This property makes a subspace robust to change in illumination condition in applications such as face recognition [13] and/or when few data is available [14]. (2) The subspaces are also statistically robust to input noises, e.g., occlusion in an image. (3) In addition, a subspace is a compact representation of a set; we can describe a set by fixed and low dimensional subspace no matter how many image frames are included in a set.

In the subspace-based methods, for classifying sets into class categories, an input subspace is compared to *dictionary subspaces* by using a structural similarity based on *canonical angles* [15,16] between subspaces. For example, suppose we have image sets of 3D objects, and it can be observed that under some assumptions, the structural similarity is related to 3D shape similarity [2–4],

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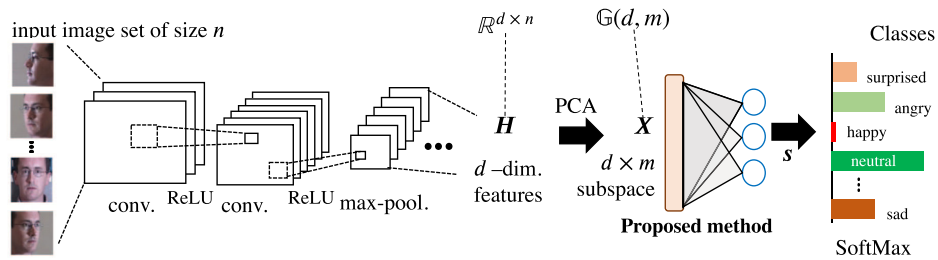


Fig. 1. Diagram of the proposed method embedded as a layer in an end-to-end learning framework.

which enables us to compare object shapes implicitly by employing subspaces.

To further improve subspace-based set classification performance, various methods of optimizing dictionary subspaces with respect to structural similarity have been proposed in the subspace literature. These methods can be divided into *analytic* and *iterative* methods. Analytic methods improve the dictionary subspaces by projecting them into an optimized feature space [12]. Seminal examples are the constrained mutual subspace method (CMSM) [14,17] and orthogonal MSM [18], which explore *linear projection* from the original feature space to the space that is contributive to discriminatively measuring the structural similarities. Although the linear projection is extended to a non-linear one via kernel tricks [19–21], the space that those methods produce is highly dependent on the input feature space in which image frames are originally represented. The iterative methods try to optimize the dictionary subspaces directly; the seminal work is performed by Kohonen and Oja [22,10] to formulate the learning subspace methods (LSM). In the LSM [22], a class dictionary subspace is iteratively improved through rotation based on the statistical decision-theoretic rule, beyond the simple SVD which computes dictionary subspace in a generative way. LSM is extended to averaged LSM (ALSM) [23] toward learning stability by considering iterative updating based on *mini-batch*. These two types of methods are developed in the subspace literature rather separately, although they could complement each other since they seek the same objective.

In this paper, we formulate a new set-classification framework (Fig. 1) to learn *both* the dictionary subspaces and the feature space (representation) in which the set subspaces are extracted.

The proposed framework contains a key module which we propose to learn dictionary subspaces and is conceptualized in Fig. 2. As in previous mutual subspace method, an input subspace is classified based on matching to class dictionary subspace by using canonical angle, a structural similarity. It should be noted that the dictionary subspaces are learned by means of the gradient descent and thereby it is possible to embed the learning subspace functionality into an end-to-end learning scheme to simultaneously learn image feature representation beyond the naive linear projection of off-the-shelf features [14,21].

From the viewpoint of learning subspace, the proposed method is connected to LSM. There, however, are several key differences between those two methods. The LSM improves a dictionary subspace in an instance-based manner by considering vector-to-subspace classification, while our method learns subspace based on subspace-to-subspace comparison for set classification. Besides, in contrast to the heuristic updating rule in LSM, we apply gradient-based update to optimize the dictionary subspaces.

Vector-to-subspace comparison in LSM is simply defined in Euclidean feature space. On the other hand, subspace-to-subspace relationships are naturally formulated in Grassmann manifold [24–26], a non-linear Riemannian manifold beyond the

Euclidean space, thereby imposing difficulty on the gradient-based updating. While the gradient-based optimization has been usually applied in Euclidean parameter space such as by Stochastic gradient descent (SGD), we establish the subspace updating formula by applying Riemannian stochastic gradient descent (RSGD) [27,28] to naturally optimize a subspace along a geodesic path towards (local) minima. It endows the learning subspace with an elegant geometrical structure of subspace as well as a clear and strong theoretical background and effective optimization; our method directly updates subspace representation without ad hoc post-processing such as SVD to ensure orthonormality of the subspace basis which is time-consuming and degrades optimality of the updating.

The proposed method to learn subspaces can thus be plugged into an end-to-end learning framework which optimizes neural networks through back-propagation, as shown in Fig. 1. As a result, the method contributes to learn both dictionary subspaces and feature representation in which the subspaces are extracted. The proposed set classification is formulated in such a general form that (1) it can cope with pre-designed (off-the-shelf) features to only learn dictionary subspaces and (2) can be embedded in an *intermediate* layer to represent an input set by a fixed-dimensional vector as set representation learning; from that viewpoint, the above mentioned set classification (Fig. 1) is regarded as a set-based classification layer whose counterpart in a standard CNN is implemented in fully-connected (FC) layer.

Our contributions are summarized as follows.

- We propose two new subspace-based methods named learning mutual subspace method (LMSM) and Grassmannian learning mutual subspace method (G-LMSM). They match subspaces and learn dictionary subspaces: LMSM learns on the Euclidean space, whereas G-LMSM learns on the Grassmannian.
- We combine LMSM and G-LMSM with CNNs, by employing CNN as backbone feature extractor and our methods as a classifier, yielding a straightforward yet powerful model for image set recognition. We showcase this combination by training models in an end-to-end manner and applying them to the problems of hand shape, face recognition, and face expression recognition from video data.
- We also propose variations of (1) a regularization loss to enhance the G-LMSM representativity; (2) a square root activation function to reprimand overconfidence; (3) fully-connected layer after a G-LMSM layer to represent more abstract objects and (4) softmax temperature to control model confidence and properly scale the logit similarity values.

2. Related works

In this section, we briefly review the recent literature of image set recognition and subspace-based learning.

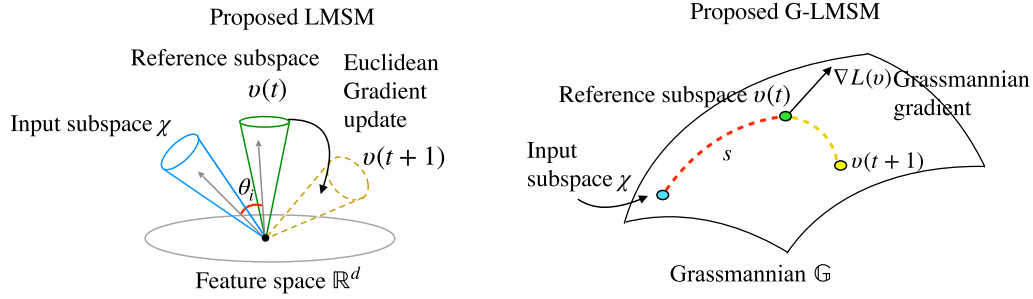


Fig. 2. Conceptualized representation of the key idea of proposed methods: learning mutual subspace method (LMSM) and learning mutual subspace method (G-LMSM). Both proposed methods match an input subspace χ (blue) to a dictionary subspace $v_j(t)$ (green) by using a similarity s_j (red) based on their canonical angles. In the learning step, LMSM updates the dictionary subspaces to $v_j(t+1)$ (yellow) using a gradient vector; LMSM uses a gradient in the Euclidean matrix space and G-LMSM uses the gradient $\nabla L(v_j)$ (black) in the Grassmannian $\mathbb{G}$.

2.1. Image set recognition

The task of image set recognition aims to recognize an object from an image set, classifying the set in one of a finite amount of defined categories. It has attracted extensive attention of research community [29–34], and various applications have found success in this paradigm, such as face recognition [35–37] and object recognition [38–40].

Zhao et al. [41] categorized the methods that approach image set recognition into parametric and non-parametric models. Parametric models, e.g. manifold density divergence [42], model sets and compare them by probability distributions and their divergences. Non-parametric models, however, do not assume a specific distribution and instead model the sets by a geometric object. The non-parametric models can be divided into: (1) linear subspace methods, such as the mutual subspace method (MSM) [43], discriminant correlation analysis (DCC) [44], constrained MSM (CMSM) [14,45]. (2) nonlinear manifold methods, such as the manifold-manifold distance (MMD) [46] and manifold discriminant analysis (MDA) [47]; and (3) affine subspace methods, e.g. affine hull based image set distance (AHISD) [48] and sparse approximated nearest point (SANP) [49,50].

2.2. Subspace-based learning

As seen in the previous section, linear subspaces can be employed as models for image sets. Although subspace-based learning intersects with the area of image set recognition, it also intersects a wide range of other fields, as many problems involve some kind of subspace structure, orthogonality or low-rank constraints, or subspace distances (e.g., canonical angles or projection norms). In most cases, the problems mathematical characteristics are expressed naturally using the Grassmann manifold. In the last few years, there have been growing interest in studying the Grassmann manifold to tackle new learning problems in computer vision, natural language processing, wireless communications, and statistical learning. Some of the recent approaches of Grassmannian learning involve metric learning, Grassmann kernels, dictionary learning and capsules. We introduce some of these works below.

Dictionary learning: Harandi et al. [51] proposes an approach based on sparse dictionary learning to learn linear subspaces, exploiting the Grassmann geometry to update the dictionary and handle non-linearity. Experiments on various classification tasks such as face recognition, action recognition, dynamic texture classification display advantages in discrimination accuracy compared to related methods.

Another method for face recognition based on dictionary learning and subspace learning (DLSL) is introduced in [52]. This new approach efficiently handles corrupted data, including noise or face

variations (e.g., occlusion and significant pose variation). The DLSL uses a new subspace learning algorithm with sparse and low-rank constraints. Results obtained through experiments on FRGC, LFW, CVL, Yale B, and AR face databases reveal that DLSL achieves better performance than many state-of-the-art algorithms.

Grassmannian kernels: Encouraged by the advantages of linear subspace representation, a discriminant learning framework has been proposed by Hamm et al. [20,53]. In this method, Various Grassmann kernel functions are developed that can map a subspace to a vector in a kernel space isometric to the Grassmannian. The subspaces are handled as a point in this kernel space through the use of a kernel trick, allowing feature extraction and classification. In the paper, kernel discriminant analysis is applied to the subspaces, a method called Grassmann discriminant analysis (GDA). In addition, experimental results on diverse datasets confirm that the proposed method provides competitive performance compared with state-of-the-art solutions.

In [54], a Grassmannian kernel based on the canonical correlation between subspaces is proposed, improving the discrimination accuracy by estimating the local structure of the learning sets. More precisely, within-class and between-class similarity graphs are generated to describe intra-class compactness and inter-class separability. Experimental results obtained on PIE, BANCA, MoBo, and ETH-80 datasets confirm that the discriminant approach improves the accuracy compared to current methods.

Another kernel-based method [55] argues that Grassmann discriminant analysis (GDA) has a decrease in its discriminant ability when there are large overlaps between the subspaces. The enhanced GDA is proposed to resolve this issue, where class subspaces are projected onto a generalized difference subspace (GDS) before mapping them onto the manifold. In general terms, GDS removes the overlapping between the class subspaces, assisting the feature extraction and image set classification conducted by GDA. Hand shape and CMU face databases are employed to show the advantages of the proposed eGDA in terms of classification accuracy.

Metric learning: Zhu et al. [56] employs nonlinear Riemannian manifolds for representing data as points, a practice widely adopted in the computer vision community. This work proposes a generalized and efficient Riemannian manifold metric learning (RMML), a general metric learning technique that can be applied to a large class of nonlinear manifolds. The RMML optimization process minimizes the geodesic distance of similar points and maximizes the geodesic distance of dissimilar ones on nonlinear manifolds. As a result, this procedure produces a closed-form solution and high efficiency. Experiments were performed using several computer vision tasks. Experimental results The experimental results show that RMML outperforms related methods in terms of accuracy.

Sogi et al. [57] propose a metric learning approach for image set recognition. The main task is to provide a reliable metric for subspace representation. The objective of the proposed metric learning approach is to learn a general scalar product space that provides improved canonical angles to estimate the similarity between subspaces. In addition, they present a formulation for dimensionality reduction by imposing a low-rank constraint. Experimental results are provided using video-based face recognition, multi-view object recognition, and action recognition datasets.

Luo et al. [58] addresses the problem that current metric learning solutions employ the L2-norm to estimate the similarity between data points. The authors argue that although practical, this approach may fail when handling noisy data. Then, they introduce a robust formulation of metric learning to solve this problem, where a low-dimensional space is utilized for producing a suitable error estimation. In addition to a robust framework, the authors present generalization guarantees by developing the generalization bounds employing the U-statistics. The proposed framework is evaluated using six benchmark datasets, obtaining high discrimination accuracy compared to related methods.

Capsules: Capsule projection (CapPro), proposed recently by [59], is a neural network layer constructed by subspace capsules. In CapPro subspaces have been incorporated in neural networks, concretely as parameters of capsules. The general idea of capsule was first proposed by [60,61], and since then, much effort has been made to seek more effective capsule structures as the base for the next generation of deep network architectures. Capsules are groups of neurons that can learn an object's variations, such as deformations and poses effectively. CapPro obtains an output vector by performing an orthogonal projection of feature vectors $\mathbf{x}$ onto the capsule subspace. The intuition behind CapPro is that it learns a set of such capsule subspaces, each representing a class. Note that CapPro cannot handle a set in a natural manner, and it does not perform subspace matching. Our proposed models can be seen as extensions of CapPro, in the sense that their forward map correspond to CapPro in the case the input consists of a single vector $\mathbf{x} \in \mathbb{R}^d$.

It is worth noting that deep learning techniques have been applied in several fields. For instance, a deep neural network for binary linear codes was recently introduced [62]. It employs the concept of deep unfolding, and two variants were analyzed. Experiments indicate that the network outperforms the traditional penalized decoder for various low-density parity-check codes with comparable computational complexity. Another example is the application of deep learning to support an intelligent reflecting surface [63]. The combination of intelligent reflecting surface and deep learning enhances the B5G wireless communications scope. Experiments indicate that the proposed method can obtain more accurate results with lower complexity than the benchmark schemes.

3. Method

We first introduce the proposed framework of learning mutual subspace method (LMSM) to classify sets and then propose Grassmannian-LMSM (G-LMSM) which optimizes subspaces in a sophisticated way based on a differential geometry of set representation.

3.1. Problem formulation

Suppose we have a set $\mathcal{H} = \{\mathbf{z}_i\}_{i=1}^n$ composed of n feature vectors $\mathbf{z}_i \in \mathbb{R}^d$, where the feature vector is extracted such as from an image by flattening raw pixels or a CNN encoder (Fig. 3). In

training, a set $\mathcal{H}$ is equipped with a class label $y \in \{1, \dots, C\}$ to construct a training dataset $\{\mathcal{H}_i, y_i\}_{i=1}^N$. It is noteworthy that the cardinality $|\mathcal{H}_i| = n_i$ can vary across samples, making it possible to flexibly describe each input set by using any number of features.

We model a set $\mathcal{H}$ by means of a subspace which geometrically represents the feature distribution $\{\mathbf{z}_i\}_{i=1}^n$. The subspace is simply computed via singular value decomposition (SVD) of $\{\mathbf{z}_i\}_{i=1}^n$ as

$$\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_n] = \mathbf{U}\mathbf{\Lambda}\mathbf{V}^T \text{ (SVD)} \Rightarrow \mathbf{Z} \approx \mathbf{U}_{1:m}\mathbf{\Lambda}_{1:m}\mathbf{V}_{1:m}^T = \mathbf{X}\mathbf{\Lambda}_{1:m}\mathbf{V}_{1:m}^T, \quad (1)$$

where the column concatenation of the n vectors is employed for producing the feature matrix $\mathbf{Z}$ and we consider the m -largest singular values to provide the basis matrix $\mathbf{U}_{1:m} = \mathbf{X} \in \mathbb{S}(d, m)$; the set of orthonormal matrices in $\mathbb{R}^{d \times m}$ is denoted by $\mathbb{S}(d, m)$, called the compact Stiefel manifold [64]. From the viewpoint of differential geometry, the subspace representation is sophisticatedly described by Grassmann manifold [64]. Therefore, we can introduce Grassmannian $\mathbb{G}(d, m)$ of $m(d - m)$ -dimensional manifold in $\mathbb{R}^d$ on which the basis matrix $\mathbf{X}$ is associated with a point $\chi \in \mathbb{G}(d, m)$. This Grassmannian representation is a foundation of the proposed G-LMSM in Section 3.3.

3.2. Learning mutual subspace method (LMSM)

It is computationally equivalent to SVD of $\mathbf{X}^T \mathbf{V}$ [64] where $\mathbf{X}$ and $\mathbf{V}$ are basis matrices of subspaces χ and ν , respectively; $\cos^2 \theta_i$ corresponds to the i -th singular value $\lambda_i(\mathbf{X}^T \mathbf{V})$. Based on this approach, we can leverage the spectral definition of the Frobenius norm to rewrite the similarity (2) as

$$s(\chi, \nu) = \sum_{i=1}^{\min(m,p)} \cos^2 \theta_i, \quad (2)$$

where the i -th canonical angle $0 \leq \theta_i \leq \frac{\pi}{2}$ between χ and ν is recursively defined by using the canonical vectors $\mathbf{u}_i$ and $\mathbf{w}_i$ [15,16];

$$\cos \theta_i = \max_{\mathbf{u} \in \chi} \max_{\mathbf{w} \in \nu} \mathbf{u}^T \mathbf{w} = \mathbf{u}_i^T \mathbf{w}_i, \quad (3)$$

$$\text{s.t. } \|\mathbf{u}_i\|_2 = \|\mathbf{w}_i\|_2 = 1, \mathbf{u}_i^T \mathbf{u}_j = \mathbf{w}_i^T \mathbf{w}_j = 0, i \neq j.$$

It is computationally equivalent to SVD of $\mathbf{X}^T \mathbf{V}$ [64] where $\mathbf{X}$ and $\mathbf{V}$ are basis matrices of subspaces χ and ν , respectively; $\cos^2 \theta_i$ corresponds to the i -th singular value $\lambda_i(\mathbf{X}^T \mathbf{V})$. Based on this approach, we can leverage the spectral definition of the Frobenius norm to rewrite the similarity (2) as

$$s(\mathbf{X}, \mathbf{V}) = \sum_{i=1}^{\min(m,p)} \lambda_i^2(\mathbf{X}^T \mathbf{V}) = \|\mathbf{X}^T \mathbf{V}\|_F^2 = \text{tr}(\mathbf{X}^T \mathbf{V} \mathbf{V}^T \mathbf{X}). \quad (4)$$

Set descriptor: An input subspace χ is described by using K dictionary subspaces' bases $\{\mathbf{V}_k\}_{k=1}^K$ into the K -dimensional vector of

$$\mathbf{s} = \begin{bmatrix} \text{tr}(\mathbf{X}^T \mathbf{V}_1 \mathbf{V}_1^T \mathbf{X}) \\ \vdots \\ \text{tr}(\mathbf{X}^T \mathbf{V}_K \mathbf{V}_K^T \mathbf{X}) \end{bmatrix} \in \mathbb{R}^K. \quad (5)$$

The descriptor can be followed by (non-linear) activation functions in neural network literature, and we apply the following function;

$$\hat{s}_j = \sqrt{s_j}. \quad (6)$$

The square root function works to suppress intense activations (s_j) while enhancing lower activations in order to properly regularize networks [65].

Learning: LMSM has a set of learnable parameters composed of K dictionary subspaces which are represented as basis matrices

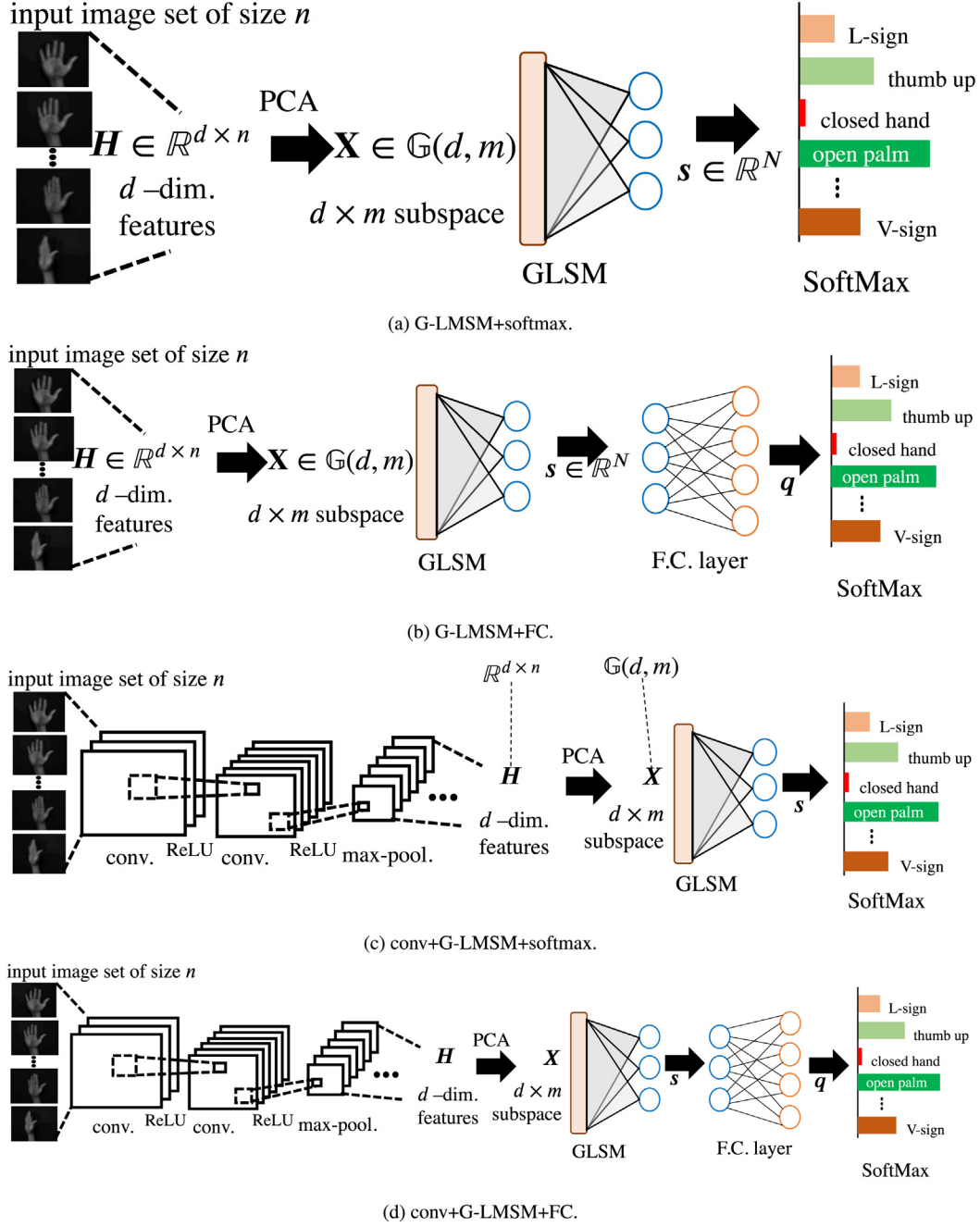


Fig. 3. Conceptual diagrams of the proposed G-LMSM architectures.

$\{\mathbf{V}_k \in \mathbb{S}(d, p)\}_{k=1}^K$. Those subspaces are initialized as random orthogonal matrices, by the following random sampling procedure: first, we sample a matrix $\mathbf{W} \in \mathbb{R}^{d \times p}$ with random Gaussian entries, i.e. $(\mathbf{W})_{ij} \sim \mathcal{N}(0, 1)$. Then we orthogonalize the matrix using QR decomposition as $\mathbf{V}\mathbf{R} = \mathbf{W}$, where $\mathbf{R}$ is a triangular matrix and $\mathbf{V} \in \mathbb{S}(d, p)$ is our sampled orthogonal matrix.

Then, the randomly initialized subspaces are optimized via back-propagation based on (softmax cross-entropy) loss L by exploiting the gradients w.r.t $\mathbf{V}_k$ as:

$$\frac{\partial L}{\partial \mathbf{V}_k} = \frac{\partial L}{\partial s_k} \frac{\partial s_k}{\partial \mathbf{V}_k} = 2 \frac{\partial L}{\partial s_k} \mathbf{X} \mathbf{X}^\top \mathbf{V}_k. \quad (7)$$

The basis matrix $\mathbf{V}_k$, however, is subject to the constraint of orthonormality $\mathbf{V}_k^\top \mathbf{V}_k = \mathbf{I}$ which has been addressed in an ad hoc

post-processing such as Gram-Schmidt orthogonalization though impeding the training. For facilitating the training, we re-parameterize the dictionary subspaces by using *unconstrained* matrix $\hat{\mathbf{V}}_k \in \mathbb{R}^{d \times p}$ as

$$s_k = \text{tr}\left\{\mathbf{X}^\top \hat{\mathbf{V}}_k (\hat{\mathbf{V}}_k^\top \hat{\mathbf{V}}_k + \epsilon \mathbf{I})^{-1} \hat{\mathbf{V}}_k^\top \mathbf{X}\right\}, \quad (8)$$

where ϵ is a small fraction parameter to numerically stabilize the inverse computation with the identity matrix $\mathbf{I}$. It is easy to verify that $\hat{\mathbf{V}}_k (\hat{\mathbf{V}}_k^\top \hat{\mathbf{V}}_k + \epsilon \mathbf{I})^{-1} \hat{\mathbf{V}}_k^\top = \mathbf{V}_k \mathbf{V}_k^\top$ where $\hat{\mathbf{V}}_k = \mathbf{V}_k \mathbf{R}$ using a regular matrix $\mathbf{R}$.

On the other hand, the following gradient w.r.t an input $\mathbf{X}$ is utilized in end-to-end back-propagation;

$$\frac{\partial L}{\partial \mathbf{X}} = 2 \frac{\partial L}{\partial \mathbf{S}_k} \mathbf{V}_k \mathbf{V}_k^\top \mathbf{X} = 2 \frac{\partial L}{\partial \mathbf{S}_k} \hat{\mathbf{V}}_k (\hat{\mathbf{V}}_k^\top \hat{\mathbf{V}}_k + \epsilon \mathbf{I})^{-1} \hat{\mathbf{V}}_k^\top \mathbf{X}. \quad (9)$$

Note that the input subspace basis $\mathbf{X}$ is computed by SVD as a orthonormal basis.

3.3. Grassmann LSM

Though the basis matrix is re-parameterized in (8), LSM optimizes dictionary subspaces in Euclidean space, missing intrinsic structure of subspaces. Thus, we propose Grassmann LSM to perform dictionary learning on the essential parameter space, Grassmannian manifold of subspaces, via the Riemannian stochastic gradient descent (RSGD) [28,66]. This manifold-aware optimization naturally keeps the orthonormal constraint of subspace bases $\mathbf{V}_k$ while maintaining a stable and efficient learning, without demanding any heuristic tricks such as SVD post-processing nor the pseudo-inverse (8). From a geometry perspective, the method efficiently traces the optimization path on the Grassmannian manifold $\mathbb{G}(d, p)$ exploiting essential geometric structures of subspaces.

The RSGD update consists of two steps. First, the Euclidean gradient $\frac{\partial L}{\partial \mathbf{V}_k}$ in (7) is transformed to a Riemannian gradient $\pi(\frac{\partial L}{\partial \mathbf{V}_k})$ which is tangent to the Grassmannian manifold. This is computed by orthogonally projecting $\frac{\partial L}{\partial \mathbf{V}_k}$ to the horizontal space at $\mathbf{V}_k$ as

$$\pi\left(\frac{\partial L}{\partial \mathbf{V}_k}\right) = (\mathbf{I} - \mathbf{V}_k \mathbf{V}_k^\top) \frac{\partial L}{\partial \mathbf{V}_k}. \quad (10)$$

The next step is to update the dictionary subspace $\mathbf{V}_k$ via the Grassmannian exponential map;

$$\begin{aligned} \mathbf{V}'_k &= \exp_{\mathbf{V}_k} \left(\eta \frac{\partial L}{\partial \mathbf{V}_k} \right) \\ &= \text{orth} \{ \mathbf{V}_k \mathbf{Q} (\cos \Theta \lambda) \mathbf{Q}^\top + \mathbf{J} (\sin \Theta \lambda) \mathbf{Q}^\top \}, \end{aligned} \quad (11)$$

where η is a learning rate and $\pi(\frac{\partial L}{\partial \mathbf{V}_k}) = \mathbf{J} \mathbf{Q} \mathbf{Q}^\top$ is the compact singular value decomposition (SVD) of the gradient. This update can be regarded as a geodesic walk towards the opposite direction to the Riemannian gradient (direction of descent), approaching a more optimal parameter point $\mathbf{V}'_k$.

3.4. Regularization on dictionary learning

The set description (5) can be impaired by a degraded dictionary in which subspaces $\{\mathbf{V}_k\}_{k=1}^K$ are highly overlapped. Such a degradation could happen when the within-category distribution is very large or the dimension $\min(m, p)$ is large compared to d . To address the issue, we propose a regularization loss to introduce repulsion among the dictionary subspaces. The proposed regularization loss is defined by

$$L_{RP} = \frac{1}{K^2} \sum_{i=1}^K \sum_{j=1}^K \frac{1}{p} \left(\|\mathbf{V}_i^\top \mathbf{V}_j\|_F^2 - \delta_{i=j} \right)^2, \quad (12)$$

where $\delta_{i=j}$ is an indicator function. The regularization loss encourages orthogonality among subspaces by letting the dictionary subspaces repel each other, so that the set descriptor (5) captures diversity characteristics of the input $\mathbf{X}$. The repulsion loss (12) is added to the objective loss (softmax classification loss) with a regularization parameter ω .

3.5. LSM in network architecture

The proposed (G-) LSM method can be embedded into a neural network in the following two ways (Figs. 3a&3b).

LSM as Classifier (Fig. 3a): As in the classical learning subspace methods, LSM is directly applied to classify an input subspace $\mathbf{X}$. In this case, the dictionary subspaces are regarded as *classifier subspaces* of which number is identical to the number of classes, $K = C$.

The set descriptor (5) is directly fed into softmax loss as logits. The logits, however, are theoretically bounded in $0 \leq s_c \leq \min(m, p)$ and thus we introduce softmax temperature τ as in [67–69] to formulate the softmax loss as $\exp(\tau s_c) / \sum_{k=1}^C \exp(\tau s_k)$. Since the temperature would depend on tasks, we apply the temperature τ as a trainable parameter optimized in an end-to-end learning to provide proper classification margin; Agarwala et al. [70] offers a detailed analysis of the underpinning learning mechanisms of softmax with temperature.

LSM as Representation (Fig. 3b): The set descriptor (5) fills a gap between a (input) set of variable length $\{\mathbf{z}_i\}_i^n$ and a (output) fixed-dimensional representation $\mathbf{s} \in \mathbb{R}^K$. Thereby, the K -dimensional descriptor can be followed by a standard neural network modules such as fully-connected (FC) classifier. In this case, the size of dictionary K is a parameter determined in advance.

4. Connection to learning subspace method

Here, we discuss the connection between the proposed LSM in Section 3 and the classic learning subspace methods from an optimization viewpoint; it should be noted that there is a clear difference between those approaches that the classical subspace methods process *vectors* in contrast to ours that focuses on *sets*.

4.1. Learning subspace method

The learning subspace method (LSM), proposed by Kohonen [22], introduces concepts of learning class subspaces with an iterative algorithm. For classifying an input vector $\mathbf{x}$ of class y , the LSM classifier produces the prediction $q \in \{1, \dots, C\}$ by

$$q = \arg\max_{c \in \{1, \dots, C\}} \|\mathbf{V}_c^\top \mathbf{x}\|_2^2, \quad (13)$$

where $\{\mathbf{V}_c\}_{c=1}^C$ are basis matrices of the classifier subspaces. During training, for a sample that is correctly classified, i.e., $q = y$, the subspace $\mathbf{V}_y$ is *reinforced* by rotation towards the sample vector $\mathbf{x}$. At the t -th iteration, the rotation is written with a learning rate α by

$$\mathbf{V}_y^{(t+1)} = (\mathbf{I} + \alpha \mathbf{x} \mathbf{x}^\top) \mathbf{V}_y^{(t)}. \quad (14)$$

On the other hand, if a sample vector is misclassified, i.e., $q \neq y$, the confusing class subspace $\mathbf{V}_q$ is *punished* by rotating it away from the sample vector $\mathbf{x}$, while the correct class subspace $\mathbf{V}_y$ is rotated towards the sample vector. These updates are written using learning rates of β and γ as

$$\mathbf{V}_y^{(t+1)} = (\mathbf{I} + \beta \mathbf{x} \mathbf{x}^\top) \mathbf{V}_y^{(t)}, \quad (15)$$

$$\mathbf{V}_q^{(t+1)} = (\mathbf{I} - \gamma \mathbf{x} \mathbf{x}^\top) \mathbf{V}_q^{(t)}. \quad (16)$$

This subspace updating procedure is simple, intuitively sound, and clearly superior to a priori methods such as subspace methods [8,9] and the Bayes classifier with normality assumptions up to a certain size [10]. To further improve the sample-wise updating procedure, the averaged learning subspace method (ALSM) [23] has been proposed by introducing mini-batch based updating. Putting the updates (14)–(16) all together, the ALSM update on mini-batch of training samples is described by

$$\mathbf{V}_c^{(t+1)} = \left(\mathbf{I} + \alpha \sum_{y_i \neq q_i \wedge y_i = c} \mathbf{x}_i \mathbf{x}_i^\top + \beta \sum_{y_i \neq q_i \wedge y_i = c} \mathbf{x}_i \mathbf{x}_i^\top - \gamma \sum_{y_i \neq q_i \wedge q_i = c} \mathbf{x}_i \mathbf{x}_i^\top \right) \mathbf{V}_c^{(t)}, \quad (17)$$

which focuses on the update of the c -th class subspace $\mathbf{V}_c$. The updated $\mathbf{V}_c^{(t+1)}$ is no longer orthonormal and thus requires Gram-Schmidt orthogonalization. The updating formulation (17) is referred to as Oja's update in this paper.

4.2. LMSM update vs Oja's update

For simplicity, the learning rates in (17) are set to $\alpha = 0$ and $\beta = \gamma$ so that the subspace is updated only if $y_i \neq q_i$, resulting in

$$\begin{aligned}\mathbf{V}_c^{(t+1)} &= (\mathbf{I} + \beta \sum_i \iota(c, q_i, y_i) \mathbf{x}_i \mathbf{x}_i^\top) \mathbf{V}_c^{(t)} \\ &= \mathbf{V}_c^{(t)} + \beta \sum_i \iota(c, q_i, y_i) \mathbf{x}_i \mathbf{x}_i^\top \mathbf{V}_c^{(t)},\end{aligned}\quad (18)$$

where $\iota(c, q, y)$ is an indicator function:

$$\iota(c, q, y) = \begin{cases} 1 & y \neq q \wedge y = c \\ -1 & y \neq q \wedge q = c \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

We can analyze the indicator function as follows.

Proposition 1. Let $s_c(\mathbf{x}) = \|\mathbf{V}_c^\top \mathbf{x}\|_2^2$ and $p_c(\mathbf{x}; \tau) = \frac{\exp(\tau s_c(\mathbf{x}))}{\sum_{k=1}^c \exp(\tau s_k(\mathbf{x}))}$ be the c -th output and the softmax probability of the LSM classifier. The softmax loss is thereby defined as $L(\mathbf{x}, \mathbf{y}; \tau) = -\log p_y(\mathbf{x}; \tau)$. For $\tau \gg 1$, the indicator function corresponds to the loss gradient;

$$\tau \gg 1 \Rightarrow \iota(c, q, y) = -\frac{1}{\tau} \frac{\partial L}{\partial s_c} \quad (20)$$

Proof. The softmax function with $\tau \gg 1$ works as the *max* operator of

$$p_c(\mathbf{x}; \tau \gg 1) = \delta_{c=q} \begin{cases} 1 & c = q (= \arg\max_k s_k(\mathbf{x})) \\ 0 & \text{otherwise} \end{cases} \quad (21)$$

The loss gradient is written by

$$-\frac{\partial L}{\partial s_c} = \tau \{\delta_{c=y} - p_c(\mathbf{x}; \tau)\}, \quad (22)$$

to which we can apply (21) and (19) to obtain (20).

Whilst, the gradient-based update in our LMSM is described using (7) with a learning rate η as

$$\mathbf{V}_c^{(t+1)} = \mathbf{V}_c^{(t)} - \eta \sum_i \frac{\partial L}{\partial s_c(\mathbf{x}_i)} \mathbf{x}_i \mathbf{x}_i^\top \mathbf{V}_c^{(t)}. \quad (23)$$

Due to the above proposition, it is reduced into the Oja's update (18) when the LMSM is applied as a classifier (Section 3.5, Fig. 3a) to 1-dimensional subspace $\mathbf{X} = \mathbf{x}$ with $\tau \gg 1$ and $\eta = \frac{1}{\tau} \beta$. Thus, the Oja's update is regarded as an extreme case ($\tau \gg 1$) of the proposed LMSM which enjoys flexible loss function with proper τ to effectively optimize the dictionary subspaces; in addition, we provide a further sophisticated optimization approach based on Grassmannian geometry in G-LMSM (Section 3.3).

5. Experiments

We have conducted a wide range of experiments to showcase the ability of LMSM to learn, both as a standalone method and in combination with other deep neural networks. Next, we first introduce the network architectures of LMSM and the datasets used in the experiments. Then, we present an ablation study to check the effect of various components we proposed. We study the effect of the optimization algorithm in the learning process, i.e., whether

constraining the gradient solver to the manifold makes a difference in the learning of LMSM. Then, we confirm the robustness of LMSM across parameter settings. Next, we present visual results, by plotting the learned latent Grassmann space for an intuitive perspective on how LMSM improves the representation by learning a more separable distribution of subspaces. Finally, we demonstrate LMSM's performance by comparing it against baselines in two types of setting: one where the image set is in raw-pixel form, and LMSM acts as a standalone method; in this case, compare LMSM against manifold-based methods. The other setting is where LMSM is combined with CNNs, so that the image appearance is processed by a CNN backbone and comes to LMSM in the form of CNN features. In this setting we compare LMSM against both classic baselines (not deep, no end-to-end learning) and deep baselines (with the same scope of architecture, but without LMSM's new features).

5.1. Network Architectures

In the following experiments, we use network architectures containing the proposed methods as a layer to approach image set recognition. The architectures are different in two separate design choices: (1) choice of set features, for example, raw images or CNN features; and (2) function of G-LMSM in the pipeline as a classifier or as a feature extractor. From these options, we can obtain the following architectures, shown on Fig. 3.

G-LMSM + softmax: This is the simplest architecture, consisting of a G-LMSM layer and a softmax applied to the activation $\mathbf{s}$. In this setting, the input set is preprocessed into a subspace as follows: Let the matrix $\mathbf{H} \in \mathbb{R}^{d \times n}$ contain the set images as its columns, where $d = w * h * c$. Then, we compute a subspace from the set using noncentered PCA, i.e., $\mathbf{H}\mathbf{H}^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top$. The subspace basis $\mathbf{X} \in \mathbb{R}^{d \times m}$ consists of the m leftmost columns of $\mathbf{U}$, that is, the columns with the highest corresponding eigenvalues. The subspace basis $\mathbf{X}$ is input in the G-LMSM layer, obtaining the activation $\mathbf{s} \in \mathbb{R}^c$. Each dictionary subspace $\mathbf{V}_j$ represents one class, and the softmax of s_j corresponds to the probability of that class given the samples. This model corresponds to CapPro in the case the input consists of a single vector $\mathbf{x} \in \mathbb{R}^d$.

G-LMSM + FC: This variation extends G-LMSM to act as a feature extractor rather than simply as a classifier, by processing $\mathbf{s}$ further through a fully-connected layer before applying softmax. The number of dictionary subspaces K becomes a free parameter, and the FC weights are $\mathbf{W} \in \mathbb{R}^{K \times c}$ and $\mathbf{b} \in \mathbb{R}^K$.

CNN backbone G-LMSM + softmax: Given the set $\mathcal{H}$ as input, we process each image $\mathcal{I}_i$ through a convolutional neural network backbone f , obtaining features $\mathbf{h}_i = f(\mathcal{I}_i)$. Then, we can create $\mathbf{H} \in \mathbb{R}^{d \times n}$ with normalized $\mathbf{h}_i$ as columns. From here we either (1) perform noncentered PCA to obtain $\mathbf{X}$ or (2) approximate the subspace projection matrix by the autocorrelation matrix (AC) $\mathbf{H}\mathbf{H}^\top$. PCA is exact, but requires an SVD decomposition, which although can be processed in an end-to-end manner by utilizing the SVD gradient updates proposed by Townsend [71], can be unstable and lead to exploding gradients. The AC is not exact, as it does not yield a subspace projection matrix, but it is a reasonable approximation here since it is quite fast and stable in end-to-end processing.

In this paper, we utilize two backbones: VGG4 and ResNet18 without the last stage of FC layers. These backbones act as feature extractors, while PCA combines the features into a subspace and a G-LMSM in the last layer acts as a classifier.

CNN backbone G-LMSM + FC: This corresponds to extracting CNN features from the set images, and then apply G-LMSM + FC, where G-LMSM here functions as a feature extractor.

5.2. Datasets

We have used three datasets in the experiments. We present the dataset details and experiment settings below.

Tsukuba hand shape dataset: A hand shape recognition dataset that contains 30 hand classes $\times$ 100 subjects, each of which contains 210 hand shape images, consisting of 30 frames $\times$ 7 different viewpoints. Example images can be seen in Fig. 4. For each subject, we randomly created 6 sets with 5 frames from each viewpoint, so that each subject has 6 image sets of 35 images. In summary, there are a total of 18000 image sets in this dataset, each set containing image information from 7 camera viewpoints. In the experiments, all the images were resized to 24×24 pixels.

We evaluated the performance of the proposed G-LMSM in the classification problem of 30 types of hand shapes. We used the image sets of 70 subjects as training sets, holding a subset of 15 subjects for validation. The remaining 15 subjects were used as testing sets.

YTC dataset: stands for YouTube Celebrities dataset. A face identification dataset [72] that contains 1910 videos of 47 identities. Similarly to [73], as an image set, we used a set of face images extracted from a video by the Incremental Learning Tracker [74]. All the extracted face images were scaled to 30×30 pixels and converted to grayscale.

Three videos per each person were randomly selected as training data, and six videos per each person were randomly selected as test data. We repeated the above procedure five times and measured the average accuracy.

AFFEW Dataset: stands for Acted Facial Expression in Wild (AFFEW) [75] dataset. It dataset contains 1,345 sequences of 7 types of facial expressions acted by 330 actors in an unconstrained setting, with scenarios close to real-world, such as movies.

We follow the experiment protocol established by [76,77] to present the results on the validation set. The training videos are split into 1747 small subvideos augmenting the numbers of sets, as established by [78]. For the evaluation, each facial frame is normalized to an image of size 20×20 . For representation, following various works [79,76,80], we represent the sequences of facial expressions with linear subspaces of dimension 10, which exist on a Grassmann manifold $\mathbb{G}(400, 10)$.

5.3. Ablation study

We conducted an ablation study experiment with the Tsukuba hand shape dataset. The objective is twofold: (1) sanity check that the proposed G-LMSM can classify objects by achieving a meaningful result, and (2) investigate the effectiveness of the additional components proposed: function of G-LMSM feature extractor (FC) or classifier (softmax), nonlinearity (sqrt), and the regularization loss (reg.). We have also added the mutual subspace method

Table 1

Results of ablation study on Tsukuba hand shape dataset.

Method	Accuracy (%)
MSM (baseline)	67.10
G-LMSM + softmax	68.87
G-LMSM + FC	89.98
G-LMSM + FC (reg.)	89.52
G-LMSM + sqrt + FC	89.69

(MSM) [14] as a base reference point since it is a fundamental method of subspace-based pattern recognition.

Table 1 shows the results. G-LMSM + softmax overperforms MSM. Although G-LMSM + softmax is the simplest possible form of G-LMSM, functioning as a linear classifier, or a one-layer neural network, this result is expected, as MSM's dictionary is obtained through PCA and does not consider the class discrimination, just class representation. Then, in G-LMSM + FC, where G-LMSM functions as a feature extractor and the FC layer as a classifier, we see a 21% increase in performance. This can be attributed to the fact that with raw-pixel input, LMSM tends to output dense vectors, not so fit for a softmax output, but very useful as features for a classifier. We can also note that the regularization and nonlinearity did not increase performance, a sign that these tools are not so useful when G-LMSM is acting as a feature extractor. But as we will discuss in the next subsections, these tools can be useful when G-LMSM is acting as a classifier and equipped with a CNN backbone.

5.4. Effect of Optimization algorithm

We compare LMSM and G-LMSM to evaluate the choice of optimization algorithm, conventional SGD or Riemannian SGD. As a baseline for comparison, we also evaluate a variation of LMSM where for the input set representation $\mathbf{X}$ we use the normalized features themselves as a basis, i.e., $\mathbf{H} = \mathbf{X}$, yielding the method named "nonorthogonal matching". Note that in this case, the similarity might not correspond to the canonical angles between $\text{span}(\mathbf{H})$ and $\text{span}(\mathbf{V})$. We equip a VGG4 backbone in LMSM, G-LMSM and the baseline nonorthogonal matching. VGG4 means the 4 initial convolutional blocks from VGG architecture. All methods, including their backbones, are randomly initialized, and there is no pretraining.

Table 2 shows the results. LMSM can outperform nonorthogonal matching, indicating that the PCA input subspace representation is effective in this task, and that having a distance based on the canonical angles is important for performance. Next the Table shows that G-LMSM outperforms LMSM, indicating that

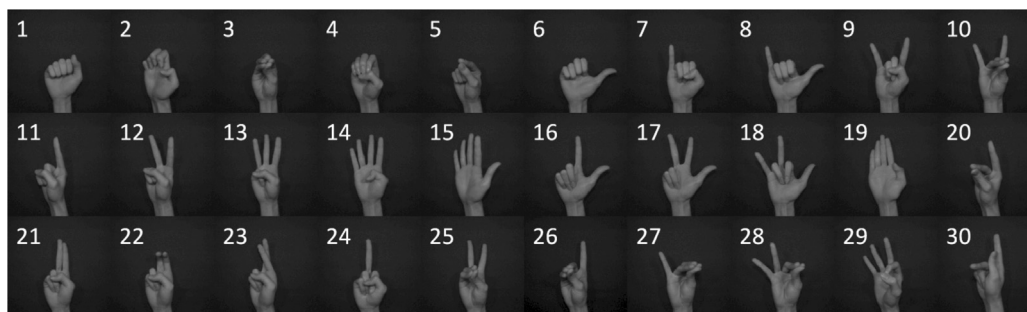


Fig. 4. Sample images from the Tsukuba hand shape dataset, displaying the 30 classes of hand shapes.

Table 2
Results of our study on Effect of Optimization algorithm.

Method	Accuracy (%)
Nonorthogonal matching	96.80
LMSM + softmax (Euclidean optim.)	98.72
G-LMSM + softmax (Grassmannian optim.)	99.39

the learning on the Grassmannian is more effective than learning on Euclidean space. The results are evidence that both the orthogonality constraint and the manifold optimization seem to play an important role in learning a subspace in an end-to-end manner.

5.5. Performance across hyperparameter settings

We conducted an experiment to assess the effects of subspace dimensions, which are hyperparameters in our model. For this experiment, we used the Tsukuba hand shape dataset. Fig. 5 shows the accuracies of G-LMSM + softmax when we vary the dimensions of subspaces. Both the dimension of dictionary subspaces p and the dimension of input subspace m are separately varied from 1 to 10. It can be inferred from the figure that G-LMSM + softmax has a lower performance from 1 to around 3 in both parameters because such low dimension subspaces are not enough to model the complexity of entities such as hand shapes, containing multiple poses, sizes, deformations and each subject's individual hand characteristics. When the dimension is 1, the G-LMSM + softmax can be regarded as a CapPro, which as seen on the figure, yields a poor performance. Increasing the dimension of the input subspace improves the performance until about $m = 8$ and $p = 7$. It can be expected that a high value of p contributes to overfitting. However, overall, it appears that the performance is not so sensitive to the value of hyperparameters if they are selected in a reasonable range, such as 5 to 10.

5.6. Visual results

To demonstrate the main point of LMSM, its end-to-end subspace learning capability, we visualize the latent Grassmannian space of two methods: mutual subspace method (MSM) and the proposed LMSM. As an image feature extraction backbone, in both cases we use a vision transformer (ViT) [81] pretrained on ImageNet. MSM is the counterpart of LMSM from a learning perspective, in that MSM does not have an end-to-end learning capability. We use ViT + MSM as a baseline to verify whether the synergy of learning ViT + LMSM makes a difference in the representation.

First, we run a first round of training of the ViT backbone for 10 epochs. This first step is done in a Euclidean manner, placing a FC layer classifier on top of the ViT to just classify the images separately without concept of image set. This allows the ViT to learn

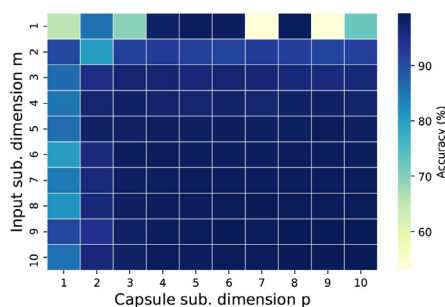


Fig. 5. Performance of G-LMSM across parameter settings, varying the dimension of capsule subspaces p and input subspaces m .

some information about the nature of our images that both MSM and LMSM can use, but it does not learn the task of image set classification. Then, in the final training stage, we replace the head/-classifier layer by either MSM or LMSM. In the case of MSM, there is no learning left, as MSM does not have the capability, whereas in the LMSM we further finetune the model to learn image set classification on LMSM while freezing the transformer blocks.

Fig. 6 shows tSNE [82,83] scatterplots of the latent Grassmannian space of both (a) MSM and (b) LMSM, showing only randomly selected 5 classes of the YTC dataset. Note that the models were trained in all 47 classes; we show a subset of 5 classes to avoid overwhelming visual pollution caused by projecting all classes in the latent space to 2D. Each dot represents a subspace in the test set; more specifically, we use each subspace similarity vector (output of either MSM or LMSM) as a representation input for tSNE, using the Grassmann kernel similarly to GDA [21]. Therefore, the plot showcases the distribution of subspaces as points in the Grassmannian generated by MSM and LMSM, so that we can compare the methods qualitatively. As we can see, the end-to-end learning of LMSM allows it to obtain very separable features compared to its counterpart without subspace learning, indicating that the proposed end-to-end subspace learning can learn useful representations.

5.7. Performance as a standalone method with raw-pixel features

We conducted an experiment to demonstrate the effectiveness of the proposed G-LMSM functioning as a standalone method. Without a CNN backbone or features, LMSM is a shallow method with only one or two layers (LMSM + FC layer), and the inputs are raw-pixel image sets. For a fair comparison within this setting, we focus on comparing shallow methods of the same scale and input type. The task is dynamic emotion recognition and the dataset is AFEW.

Baselines: As the G-LMSM is a network layer that learns subspaces based on iterative Riemannian optimization, we compare it against the following subspace/ manifold-based methods: (1) a regular CNN that can handle image sets, namely the Deep Second-order Pooling (DeepO2P) [84]. (2) Methods based on subspace without Riemannian optimization: DCC [5], Grassmann Discriminant Analysis (GDA) [20] and Grassmannian Graph-Embedding Discriminant Analysis (GGDA) [21]. And (3) methods based on Riemannian optimization: Projection Metric Learning (PML) [85], Expressionlets on Spatio-Temporal Manifold (STMExpLet) [76], Riemannian Sparse Representation combining with Manifold Learning on the manifold of SPD matrices (RSR-SPDML) [86], Network on SPD manifolds (SPDNet) [77] and Grassmann net (GrNet) [80]. Especially, GrNet proposes a block of manifold layers for subspace data. GrNet-1 denotes the architecture with 1 block and GrNet-2 with 2 blocks.

Results and discussion: The results can be seen in Table 3. The proposed method achieved quite competitive results compared to the manifold-based methods, by mapping the subspace into a vector and then uses simple Euclidean operations such as fully-connected layers and cross-entropy loss, in contrast to several methods that use complex approaches requiring multiple matrix decompositions. First, G-LMSM + softmax outperforms popular methods such as GDA and GGDA, which have a similar purpose of mapping Grassmann manifold data into a vector representation. The reason is perhaps that G-LMSM learns both the representation and discrimination in an end-to-end manner, while GDA uses a kernel function to represent subspaces and learns the discriminant independently. Methods such as SPDNet and GrNet are composed of many complex layers involving SVD, QR decompositions, and Gram-Schmidt orthogonalization and its derivatives are utilized as well. They increase in complexity as the number of layers

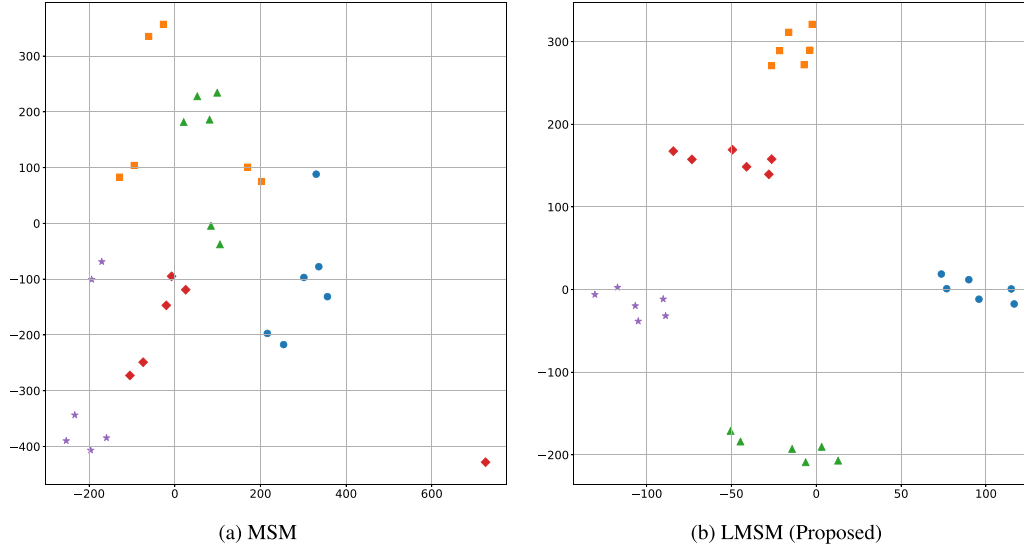


Fig. 6. tSNE scatterplot of subspaces of 5 classes of the YTC dataset. Each dot represents a subspace, so the plot showcases the distribution of subspaces as points in the Grassmannian. The left plot shows (a) MSM and the right plot (b) shows (b) LMSM (proposed). In both cases, the backbone is a vision transformer (ViT) pretrained on ImageNet. MSM is the classical counterpart of LMSM, and does not have an end-to-end capability. As we can see, the end-to-end learning of LMSM allows it to obtain very separable features.

Table 3

Results of the experiment on the AFEW dataset.

Method	Accuracy (%)
STM-ExpLet	31.73
RSR-SPDML	30.12
DCC	25.78
GDA	29.11
GGDA	29.45
PML	28.98
DeepO2P	28.54
SPDNet	34.23
GrNet	34.23
G-LMSM + FC	38.27

increases by repeating these operations, which are not easily scalable to use in GPUs. On the other hand, the proposed method provides competitive results with fewer layers and no decompositions, making it naturally parallelizable and scalable.

5.8. Performance in combination with deep backbones

In this subsection, we compare combine G-LMSM with various CNNs and compare its performance to multiple deep learning baselines to evaluate its effectiveness.

Table 4

Results of the experiment on the Tsukuba hand shape dataset.

Feature	Method	Accuracy (%)
No backbone	Baselines	MSM
		67.10
	Proposed	G-LMSM + softmax
		68.87
		G-LMSM + FC
With backbone (VGG4)	Baselines	G-LMSM + FC (reg.)
		89.98
		G-LMSM + sqrt + FC
		89.52
		89.69
	Proposed	Set average
		97.54
		Bilinear
		83.81
		Softmax average
		97.48
		CapPro average
		96.80
		Nonorth. matching
		96.80
		LMSM + softmax
		98.72
		PCA + G-LMSM + softmax
		99.39
		PCA + G-LMSM + FC
		99.06

5.8.1. Hand shape recognition

We have performed an experiment with the Tsukuba hand shape dataset.

Baselines: There are five baselines in this experiment: (1) the mutual subspace method (MSM) [14], a fundamental method of classifying subspaces with canonical angles. (2) "set average", a very simple idea to test the effectiveness of the backbone network. The set images are processed through the backbone VGG4, yielding a matrix of output features $\mathbf{H} \in \mathbb{R}^{512 \times 35}$, where the backbone features are of size 512 and the number of images is 35. Then the set images are averaged, which can be regarded as a setwise global average pooling, resulting in a single 512 vector $\mathbf{h}$, which is then processed through a FC layer and softmax. (3) Softmax average, which processes each image through a FC layer and softmax and then averages the output probabilities. (4) Bilinear [87], which uses the features correlation matrix $\mathbf{H}\mathbf{H}^T$ as a single integrated feature, and classifies it through an FC layer. And (5) CapPro average, where the average vector $\mathbf{h}$ is used as input to a CapPro layer.

Results and discussion: Table 4 shows the results. The first results, with no backbone convolutional layers, suggest two points: G-LMSM-based models can classify hand shapes more accurately than MSM, even when it is used on its own, i.e., just the G-LMSM layer as a standalone classifier; and the G-LMSM acting as a feature

extractor followed by an FC classifier offers a much better performance. Neither the square root activation nor the regularization loss seems to offer a gain of accuracy in this case.

The second part of Table 4 shows methods that perform end-to-end learning with a VGG4 backbone. These results demonstrate two points: (1) the average feature vector of a set is a naive approach to treating image sets and that more information can be modeled from the features distribution; and (2) interestingly, the G-LMSM in a deep backbone network seems to perform better as a classifier than as a feature extractor. The reason might be that either the FC layer followed G-LMSM is prone to overfitting or that, in this kind of architecture, subspaces are better at representing classes directly rather than more abstract entities.

5.8.2. Experiment on Face Identification

We conducted an experiment of face identification with the YouTube Celebrities (YTC) dataset.

Baselines: (1) Classic image set-based methods: Discriminative Canonical Correlations (DCC) [5], manifold-manifold distance (MMD) [88], Convex Hull-based Image Set Distance (CHISD) [48], Pairwise linear regression classification (PLRC) [32], image-set based collaborative representation and classification (ISCRC) [89], probabilistic collaborative representation on Grassmann manifold for image set classification (PGCRC) [90]. (2) Deep learning methods: deep reconstruction model (DRM) [91]; Resnet vote [92] is a baseline consisting of a Resnet50 [92] fine-tuned to this dataset with the cross-entropy loss in a single image setting. For the fine-tuning, we added two fully connected (FC) layers after the last global average pooling layer in the network. The first FC layer outputs a 1024 dimension vector through the ReLU [93] function, and the second layer outputs a 47 (the number of classes) dimension vector through the softmax function. Hyperparameters of the optimizer were used as suggested by the original paper. For classification, each image of a set is classified independently and a majority voting strategy of all predictions is used to select a single class prediction for the whole set. The model called Resnet18 set average is the same as used in the hand shape experiment. The Resnet18 bilinear model uses the correlation matrix of the backbone features of the image set, generating a 512×512 matrix. The vectorized matrix is processed through an FC layer and softmax for classification.

Results and discussion: Table 5 shows the results. G-LMSM outperforms not only the classical manifold methods, but also various deep methods. Although G-LMSM uses the same backbone as some methods, it still can achieve better results. In the case of Resnet50 vote and the G-LMSM + FC, the key difference is that voting is a simple heuristic approach to image set recognition, as the model is not

aware of the set as a whole. This leads to unstable and diverging predictions on a set's class, as it does not take into account the underlying set distribution. The voting method is not differentiable by nature, so it is not trivial to extend this idea to an end-to-end approach. Then, set average seems again to be a naive approach to treating image sets and that more information can be modeled from the features distribution. The bilinear model could in theory capture the underlying set information from the correlations, but the high dimensionality of the correlation matrix makes it almost impossible to learn a FC classifier without overfitting.

These results confirm the advantage of G-LMSM over these methods using the same backbone: mutual subspaces provide a low-dimensional, differentiable approach to modeling the set distribution, ultimately leading to a straightforward integration of the deep convolutional networks and the problem of image sets.

Furthermore, the AC + G-LMSM + softmax seems to yield the same result as PCA + G-LMSM + softmax, evidence that it is a good approximation for processing the input data. Then, the proposed variations of G-LMSM, namely, regularization loss, the square root activation and the temperature softmax (temp-softmax) appear to provide an advantage over the base G-LMSM.

6. Conclusion

In this paper, we addressed the problem of image set recognition. We proposed two subspace-based methods, named learning mutual subspace method (LMSM) and Grassmannian learning mutual subspace method (G-LMSM), that can recognize image sets with DNN as feature extractors in an end-to-end fashion. The proposed methods generalize the classic average learning subspace method (ALSM), in that LMSM and G-LMSM can handle an image set as input and perform subspace matching. The key idea of G-LMSM is to learn the subspaces with Riemannian stochastic gradient descent on the Grassmann manifold, ensuring each parameter is always an orthogonal basis of a subspace.

Extensive experiments on hand shape recognition, face identification, and facial emotion recognition showed that LMSM and G-LMSM have a high discriminant ability in image set recognition. It can be used easily as a standalone method or within larger DNN frameworks. The experiments also reveal that Riemannian optimization is effective in learning G-LMSM within neural networks.

Our study presents two contributions to the literature: 1) it addresses DNNs' inability to cope with image sets, an open problem in the field. 2) Our method integrates subspace-based methods into modern DNNs while keeping end-to-end trainability, which was not possible before. Especially, the proposed LMSM general-

Table 5
Results of the experiment on the YouTube Celebrities dataset.

	Type	Method	Accuracy (%)
Baselines	Classic	DCC	51.42 ± 4.95
		MMD	54.04 ± 3.69
		CHISD	60.42 ± 5.95
		PLRC	61.28 ± 6.37
		PGCRC	71.35 ± 0.09
		ISCRC	69.70 ± 7.40
	Deep	DRM	66.45 ± 5.07
		Resnet50 vote	64.18 ± 2.20
		Resnet18 set average	68.24 ± 3.32
		Resnet18 bilinear	67.85 ± 3.35
Proposed	Features (from ResNet50)	G-LMSM + FC	66.17 ± 4.19
	End-to-end (backbone ResNet18)	PCA + G-LMSM + softmax	68.80 ± 3.30
		AC + G-LMSM + softmax	68.93 ± 3.97
		AC + G-LMSM + sqrt	71.09 ± 3.62
		AC + G-LMSM (reg.)	70.01 ± 2.87
		AC + G-LMSM + temp-softmax	69.92 ± 3.79

izes the learning subspace method (LSM), a classic statistical method, by providing a theoretically well-grounded approach to bridge the gap between the classical pattern recognition approach and modern DNN.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lincon Souza reports financial support was provided by Japan Society for the Promotion of Science.

Acknowledgements

This work was supported by Japan Society for the Promotion of Science (JSPS) KAKENHI Grant No. JP22K17960.

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